like esterases, acid phosphatases as well as compatibility controlling proteins.

Solid style (Closed type): It is common among dicots. It is characterized by the presence of central core of elongated, highly specialised cells called transmitting tissue.This is equivalent to the lining cells of hollow style and does the same function. Its contents are also similar to the content of those cells. The pollen tube grows through the intercellular spaces of the transmitting tissue.

Semi-solid style (half closed type): This is intermediate between solid and open type.

There is a difference of opinion on the nature of transmitting tissue. Some authors consider that it is found only in solid styles while others consider the lining cells of hollow style also has transmitting tissue.

Entry of pollen tube into the ovule: There are three types of pollen tube entry into the ovule(Figure 1.18).

a) Porogamy: when the pollen tube enters through the micropyle.

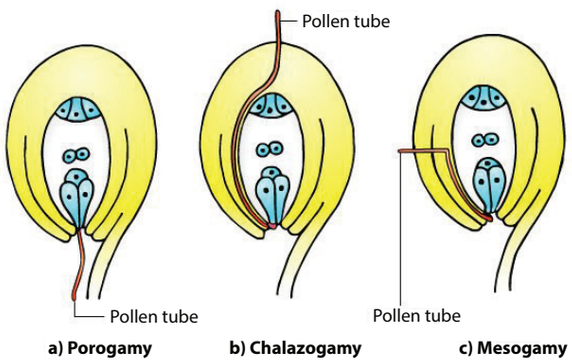


Figure 1.18 Path of pollen tube entry into the ovule

b) Chalazogamy: when the pollen tube enters through the chalaza.

c) Mesogamy: when the pollen tube enters through the integument.

Entry of pollen tube into embryo sac: Irrespective of the place of entry of pollen tube into ovule, it enters the embryo sac at the micropylar end. The pollen enters into embryo sac directly into one of the synergids.

The growth of pollen tube towards the ovary, ovule and embryo sac is due to the presence of chemotropic substances. The pollen tube after travelling the whole length of the style enters into the ovary locule where it is guided towards the micropyle of the ovule by a structure called **obturator** (See Do you know). After reaching the embryo sac, a pore is formed in pollen tube wall at its apex or just behind the apex. The content of the pollen tube (two male gametes, vegetative nucleus and cytoplasm) are discharged into the synergids into which pollen tube enters. The pollen tube does not grow beyond it, in the embryo sac. The tube nucleus disorganizes.



Obturator may originate from different regions of the ovule (Placenta – Euphorbiaceae, Funiculus-Anacardiaceae, Style–*Thymelaeaceae* and Ovary wall – *Ottelia alismoides*)

1.5.1 Double fertilization and triple fusion

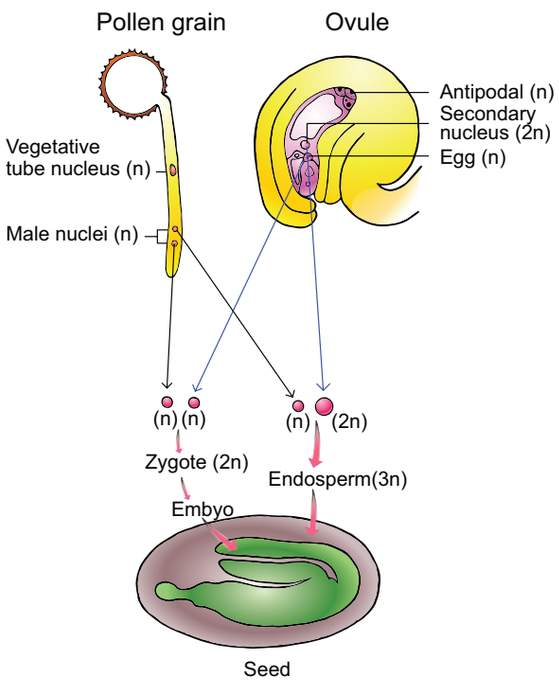


Figure 1.19 Fertilization in Angiosperms

S.G. Nawaschin and L.Guignard in 1898 and 1899, observed in *Lilium* and *Fritillaria* that

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